

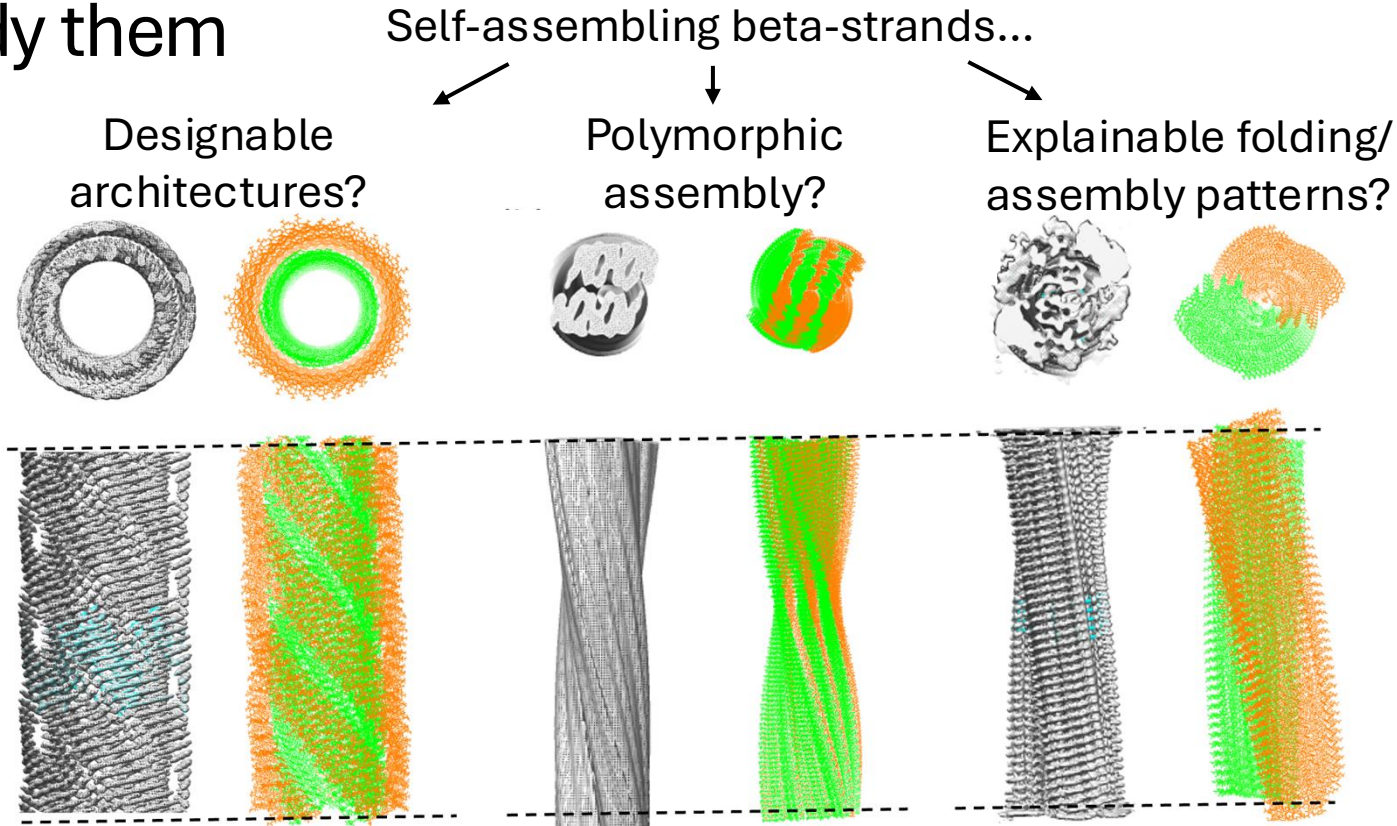
2026 Building Molecular Systems Workshop

FibrilGen: a Python package to generate cross-beta nanostructures

Speaker: Chao-Yu Yang

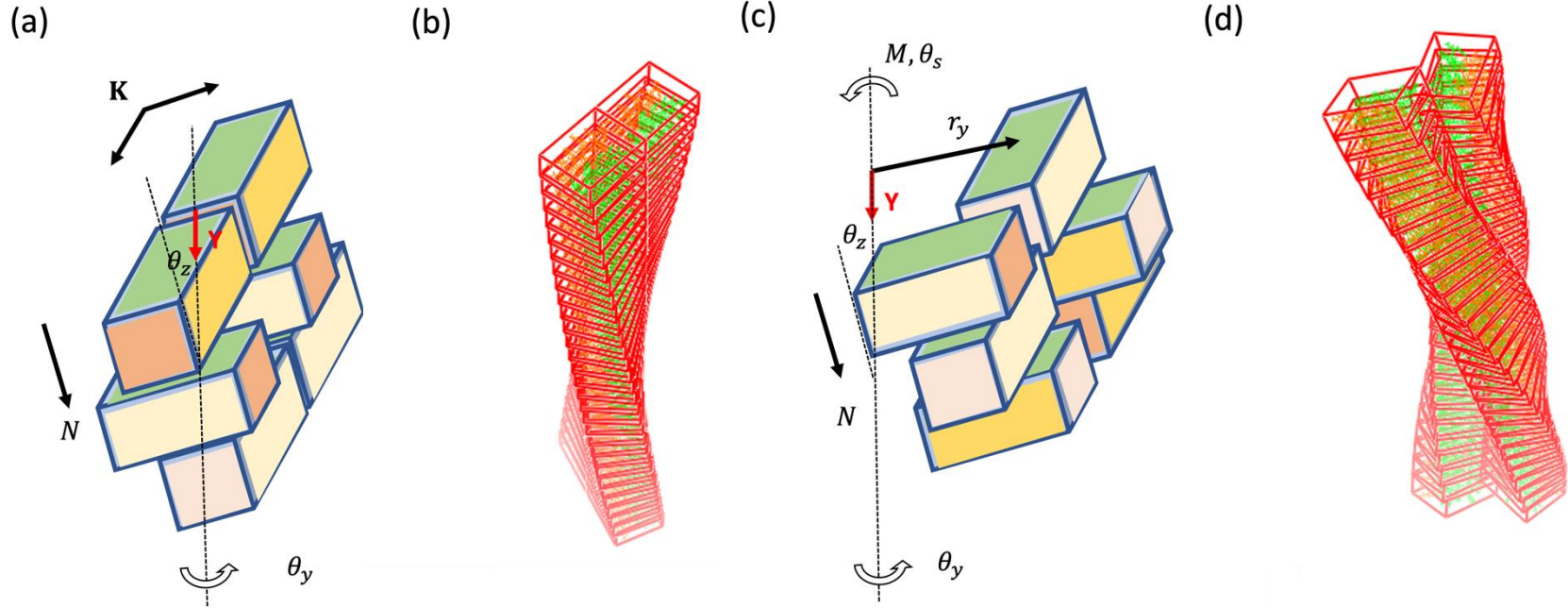
Date: 27/11/2026

What are cross-beta nanostructures and why we want to study them

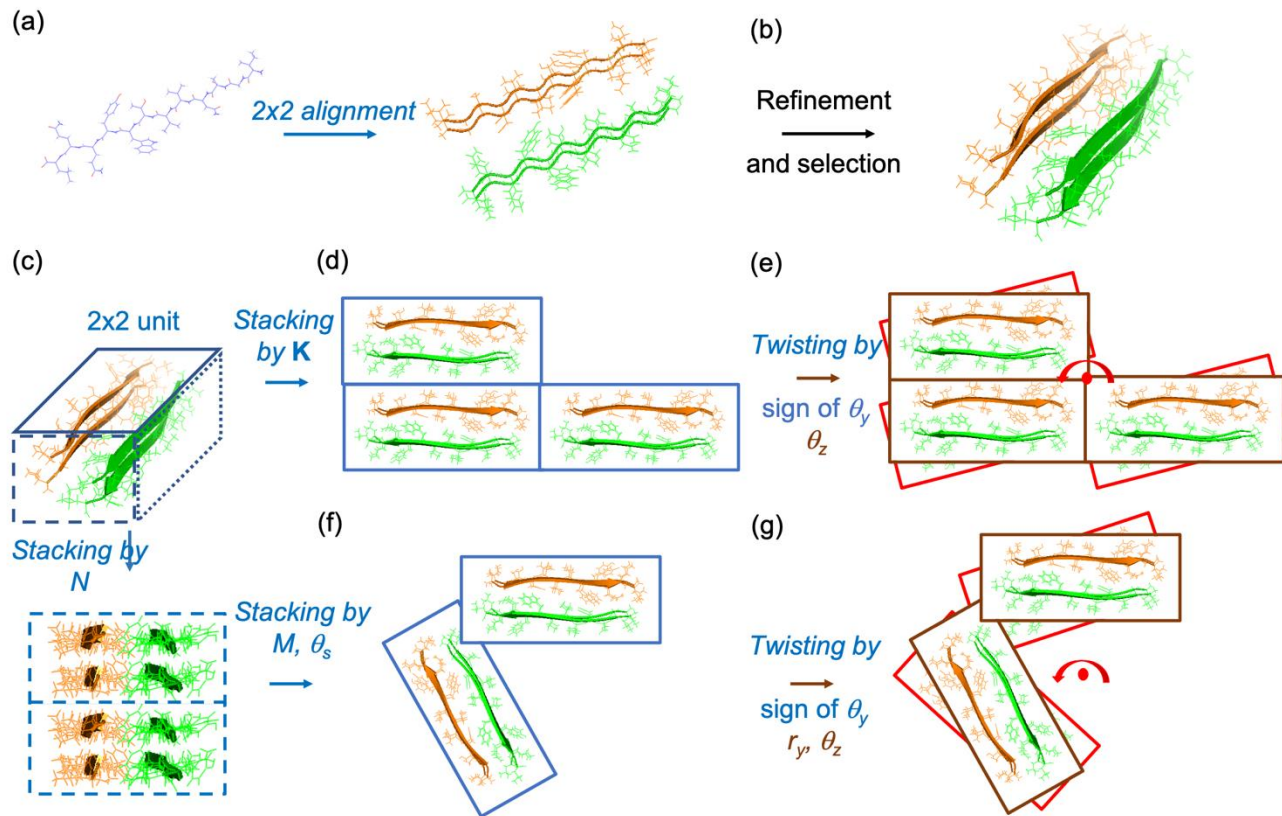


Wang, F. et al. (2021), Schmidt, A. et al. (2016), Gremer, L. et al. (2017), Yang, C.-Y. et al. (2025)

An overview of common cross-beta geometry

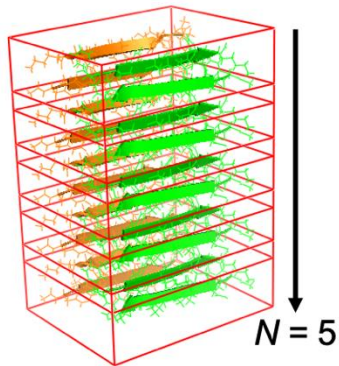


Assemble a peptide molecule into a cross-beta structure



Seven parameters for six classes of cross-beta models

Class 1



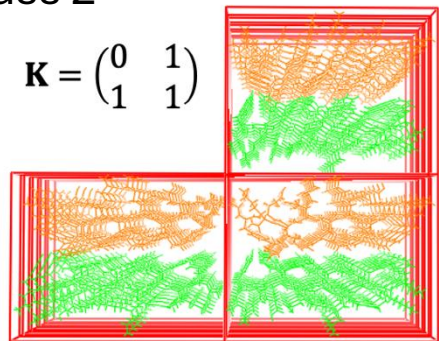
Class 3 $\theta_y = 4.1$



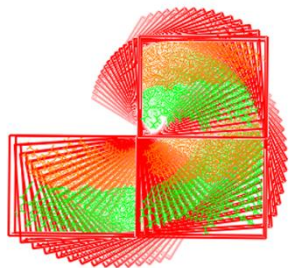
Class 5 $r_y = 30$



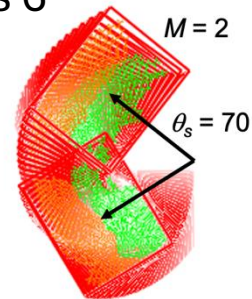
Class 2



Class 4



Class 6



Let us build cross-beta structures!



Build a beta-strand

Basic

Input: a peptide sequence

Output: a beta-strand

Advanced

How do side-chain rotations create steric clashes?



Practice

1. Try a different peptide sequence
2. Try a different side-chain rotation angle



Build a bilayer structure

Basic

Input: a peptide molecule, peptide alignment for a 2 x 2 unit

Output: an energy minimised beta-sheet

Advanced

How peptide molecule and peptide alignment destabilise the beta-sheet assembly?



Practice

1. Try a different peptide molecule
2. Try a different peptide alignment



Build a fibril structure

Basic

Input: a 2 x 2 unit, input parameters for a fibril model

Output: a cross-beta structure

Advanced

How do 2 x 2 units affect the feasible assembly of beta-sheet structure?



Practice

1. Try a different N . Increase the number for a longer fibril.
2. Try a different K . Increase the number of presented units for a thicker fibril.
3. Try a different sign of θ_y . Assign 1 for a left-handed twist and -1 for a right-handed twist.
4. Try a different θ_z . Increase the number for a shorter fibril.
5. Try a different r_y . Increase the number for a wider fibril.
6. Try a different M . Increase the number of presented units for a more enclosed fibril.
7. Try a different θ_s . Increase the number for a more twisted fibril.

We welcome feedback and suggestions!